THE EULER CHARACTERISTIC OF THE FIBRES OF A PROPER FLAT MORPHISM IS LOCALLY CONSTANT

THE STACKS PROJECT AUTHORS

Editorial note. Inputs lemma-flat-proper-perfect-direct-image-general and lemma-chi-locally-constant are same-chapter NOT carded (cross-line-claimed labels) and are declared external.

Allowed standard prerequisites (editorial).

- Perfectness and base change compatibility of the derived pushforward. lemma-flat-proper-perfect-direct-image-general (same chapter, NOT carded, declared external). K = Rf_*F is perfect with formation commuting with arbitrary base change.
- Alternating sum of fibre cohomology dimensions is locally constant. lemma-chi-locally-constant (same chapter, NOT carded, declared external). The displayed function s -> sum (-1)^i dim H^i(K tensor kappa(s)) is locally constant.

Source reference. Original lemma 0BDJ: sources/stacks-perfect/pinned-original-source.tex, source lines 7995–7995. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0B91: sources/stacks-perfect/pinned-original-source.tex, source lines 7742–7742. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a proper morphism of finite presentation. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module of finite presentation, flat over S. The function*

$$s \longmapsto \chi(X_s, \mathcal{F}_s)$$

is locally constant on S. Formation of this function commutes with base change.

Proof. By cohomology and base change (more precisely by Lemma 0B91) the object $K = Rf_*\mathcal{F}$ is a perfect object of the derived category of S whose formation commutes with arbitrary base change. Thus we have to show the map

$$s \longmapsto \sum (-1)^i \dim_{\kappa(s)} H^i(K \otimes^{\mathbf{L}}_{\mathcal{O}_S} \kappa(s))$$

is locally constant on S. This is Lemma 0BDJ. □

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